Marshall Bruner

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RF hardware, signal processing, and weather radar deployment | Teaching 50k+ engineers on YouTube

EXPERIENCE

Rainmaker Technology Corporation — Aug 2025 — Oct 2026
Head Radar Systems Engineer — *Boulder, CO*

- **Led radar engineering for a cloud-seeding company**: set S-, X-, and K-band sensor strategy and owned radar hardware, FPGA / embedded software, signal processing, and deployment tooling
- **Designed a 24 GHz FMCW cloud-profiling radar transceiver** (8-layer Rogers PCB, 450+ components, KiCad) from architecture through fab and bring-up, including a bistatic redesign for 15+ dB sensitivity gain and GaN PA bias sequencing
- Owned its software stack: SystemVerilog gateware on Zynq UltraScale+ (PLL control, LVDS ADC capture, DMA), C++ control server, Python capture and Doppler processing pipeline, and SCPI-automated bench characterization
- **Operated and deployed a 25 kW X-band dual-pol scanning weather radar**: scan-control and monitoring service with Slack / SNMP alerting, remote networking, and a mountain-top installation
- Architected a solid-state pulse-compression X-band dual-pol successor radar (transceiver design, PA trade study, sensitivity budget, business case) and built terrain-aware coverage-planning and radar performance-modeling tools used to site radars in 12 program regions

CSU-CHILL National Radar Facility — Jan 2025 — Present
Radar Systems Engineer (part-time) — *Greeley, CO*

- **Built the facility's multi-radar web display** (Flask, OpenLayers, PostgreSQL, Docker) serving real-time and historical PPI / RHI views for 8 radars and 14 polarimetric products, with client-side GPU rendering and a multi-layer cache
- **Wrote the antenna positioner controller for a solid-state C-band radar** in Go: reverse-implemented the pedestal's binary UDP protocol, PPI / RHI scan planner and queue, operator web UI, and CLI
- Maintained an X-band dual-pol signal processing chain in C (time series to moments to CF-Radial NetCDF to polarimetric products) and authored its operational volume coverage patterns
- Supported relocation and re-commissioning of a dual-pol S-band weather radar at Reagan Test Site, Kwajalein Atoll: bring-up procedure, calibration verification, and a live monitoring dashboard

BAE Systems (DoD Clearance) — Jul 2024 — Jul 2025
RF Engineer — *Westminster, CO*

- **System design, schematic capture, and layout of a 2D active phased array weather radar**
- Large-scale phased array systems analysis in Python and Matlab
- Technical consultant on weather radar proposals

Ball Aerospace (DoD Clearance) — Jun 2021 — Jul 2023
RF Engineer — *Westminster, CO*

- **Designed and debugged multiple mission-critical RF boards under strict time constraints**
- Simulated, designed, and tested an RF front-end PCB at extreme temperatures
- **Schematic capture and layout of RF T/R systems** (RF, digital, power), antenna range test equipment, and component test coupons
- Built an embedded webpage used to control and monitor phased array electronics; high-sensitivity noise and high-power testing

Colorado State University **Jan 2021 — May 2022**
Graduate Research Assistant — Thesis focused on Radar *Fort Collins, CO*

- **Design, schematic, and layout** of a modular X-band FMCW drone radar
- **Signal processing and data visualization** for Analog Devices' FMCW Phaser radar board; presented by ADI at IMS 2022 and IEEE-APS 2022
- Presented the work at the IEEE International Symposium on Phased Array Systems and Technology

CSU-CHILL Radar Lab **May 2020 — May 2021**
Lab Assistant *Greeley, CO*

- Built a server to host and display real-time and historical radar data using PostgreSQL, Flask, and Python
- Performed characterization and measurements of radar equipment

PROJECTS

Marshall Bruner RF — Radar / RF Education YouTube Channel **Aug 2023 — Present**

- **53k subscribers and 1.4M+ views** across 17 long-form animated videos on FMCW radar, CFAR, phased arrays, pulse compression, I/Q sampling, and dual-pol weather radar; top video 320k+ views
- Write, animate (Manim), narrate, and edit every video; built a reusable radar animation library (WSR-88D model, PPI displays, beam geometry, NEXRAD ingest) and contributed features to the ManimGX GPU engine
- Ship companion interactive notebooks (marimo to WASM) and a Python / Three.js 3D weather-radar animation studio with DEM terrain and 4/3-Earth beam refraction

FMCW Radar Bootcamp — IEEE AESS RadarConf **Mar 2023**

- Developed **signal processing software and a presentation for an FMCW radar** for the 2023 AESS Radar Bootcamp

EDUCATION

Colorado State University **Jan 2021 — May 2022**
*M.S. Electrical Engineering — **GPA: 3.9*** *Fort Collins, CO*
Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight FMCW Radar

Colorado State University **Aug 2017 — May 2021**
*B.S. Electrical Engineering — **GPA: 3.47*** *Fort Collins, CO*

SKILLS & ABILITIES

Radar / RF: FMCW and pulsed radar design, pulse compression, dual-polarization polarimetry, phased arrays, link budgets, Doppler / spectral moment estimation, CFAR, receiver noise and sensitivity analysis, weather radar operations and deployment

Languages: Python (NumPy, SciPy, Pandas, PyTorch, FastAPI, Flask), Rust, Go, C / C++, SystemVerilog, Bash, SQL, Matlab, TypeScript

Design / Simulation Tools: KiCad, AWR Microwave Office, HFSS, ADS, LTSpice, Cadence, QucsStudio, Vivado

Lab / Infrastructure: VNA, spectrum analyzer, oscilloscope, noise figure and high-power test; Git, Linux, Docker, PostgreSQL, Tailscale, LaTeX, Manim

Communication: Technical presentations and design reviews (PDR / CDR), customer pre- and post-sales support, educational content reaching 50k+ engineers